

EduNexus Institute of Technology

End Semester Examination

ME301 - Mechanics of Vibrations ? Sample text ?????? ?? with symbols: ? A ? B, P ? Q, damping ? and ?, em?dash, café

Date: 2026-08-17 Time: 180 Minutes

Maximum Marks: 30

Instructions:

1. All questions carry marks as indicated. Sample text ?????? ?? with symbols: ? A ? B, P ? Q, damping ? and ?, em?dash, café

SECTION - A

Answer using proper notation.

Q - 1 A Sample text ?????? ?? with symbols: ? A ? B, P ? Q, damping ? and ?, [10] CO1 3
em?dash, café. Given the table below:

Trial	Load (N)	Deflection (mm)	Stress (MPa)	Remarks
1	100	0.42	12.5	within limit
2	200	0.88	25.1	within limit
3	300	1.31	37.8	within limit
4	400	1.79	50.2	near yield
5	500	2.35	62.9	yield onset
6	600	3.02	75.4	plastic
7	700	3.91	88.0	plastic
8	800	5.20	100.6	necking
9	900	7.10	113.1	necking
10	1000	9.85	125.7	fracture

Find deflection at 500N using $\frac{d^2y}{dx^2} + \omega^2 y = 0$. Attached image below shows the setup.



Q - 2

[05]

CO BTL PO

(a) Which damping ratio ζ gives fastest return without oscillation? Sample text
 ?????? with symbols: ? A ? B, P ? Q, damping ? and ?, em?dash, café

- a) < 1
- b) $= 1$
- c) > 1
- d) $= 0$

Q - 3 Tabulate the following results:

[15] CO3 4

Trial	Load (N)	Deflection (mm)	Stress (MPa)	Remarks
1	100	0.42	12.5	within limit
2	200	0.88	25.1	within limit
3	300	1.31	37.8	within limit
4	400	1.79	50.2	near yield
5	500	2.35	62.9	yield onset
6	600	3.02	75.4	plastic
7	700	3.91	88.0	plastic
8	800	5.20	100.6	necking
9	900	7.10	113.1	necking
10	1000	9.85	125.7	fracture

CO : Course Outcome Number BTL : Bloom's Taxonomy Level

1: Remember	2: Understand	3: Apply	4: Analyze	5: Evaluate	6: Create
-------------	---------------	----------	------------	-------------	-----------
